

# James Dinh

## Full Stack Data Engineer

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### Experience

#### Data Analytics & UX Intern

Aug 2025 – present

*Fung Institute for Engineering Leadership – UC Berkeley MEng Program*

*Berkeley, CA*

- ❖ Cleaned, aggregated, and analyzed student pulse-survey data using Python and automated quality checks, producing reliable program metrics that informed Capstone Project resource allocation for 450+ MEng students
  - ❖ Developed a comprehensive MAS-E course-planning tool with automated prerequisite checks and forecasting logic, and integrated its data flows with Salesforce Data Cloud to unify advising and reporting across the program
- Python (Pandas) / JavaScript (Apps Script) / Salesforce / Google Sheets*

#### Full Stack Software Engineer

Sep – Dec 2025

*Prompt Opinion – AI Healthcare Copilot Platform*

*Remote (Contract)*

- ❖ Engineered and integrated MCP tools and AI Agents in C# (.NET 9), implementing FHIR-compliant API workflows and reducing LLM-EHR data retrieval latency through optimized HTTP pipelines of patient data and Vue.js caching
  - ❖ Developed and maintained full-stack features in close collaboration with Design and Product, using Vue.js, TypeScript, Vite, Node.js, and PostgreSQL; deployed to Azure via Docker and GitHub Actions CI/CD pipelines
- Vue / TypeScript / C# .NET / PostgreSQL / Azure / Node.js / Docker*

#### AI & Data Science Intern

Jun – Aug 2025

*Darena Health – AI-enabled, FHIR-first EHR*

*Remote*

- ❖ Built an MCP server and data-integration pipeline connecting an LLM-powered clinical chatbot to an EHR, enabling real-time retrieval of patient FHIR data to power CVD risk scoring models and clinical trial matching
  - ❖ Containerized and deployed the MCP server and data-integration pipeline using Docker and Kubernetes, providing consistent, isolated environments for bidirectional FHIR data exchange and improving stability for new features
- TypeScript / Node.js / Docker / Kubernetes*

#### Frontend Web Developer Intern

Aug 2023 – Aug 2024

*Digital NEST – Web & Graphic Design Nonprofit*

*Gilroy, CA*

- ❖ Coordinated with clients in education and recreation to produce, develop, and maintain websites using WordPress, JS, and PHP, implementing SEO best practices to boost page visits by 200% and sustain an average 36% bounce rate

### Education

#### University of California, Berkeley

Expected Grad May 2026

B.A. Data Science

GPA: 3.85

- ❖ Coursework: ML, Data Engineering, Data Structures, Probability Theory, Numerical Analysis
- ❖ Concentration in Applied Math & Modeling; Clubs/Orgs: Cal VSA Lion Dance (Music Coordinator)

#### Gavilan College, Dual Enrollment with High School

Grad May 2023

A.S. Computer Programming | A.A. Natural Science

GPA: 3.95

### Projects

#### MIMIC-IV-Ext Unsupervised Phenotyping of Diabetes Progression (Research Data Scientist)

Data Pipeline Utilizing K-Means & UMAP to Phenotype Diabetic Patients – [Research Paper](#)

- ❖ Engineered a scalable preprocessing and feature engineering pipeline to evaluate unsupervised segmentation models on 46k+ patients, identifying clinically meaningful clusters to support downstream risk analysis
  - ❖ Trained a PyTorch transformer to learn patient-level embeddings from heterogeneous, temporally irregular data
- Python (Pandas, Scikit-learn, PyTorch, Seaborn) / Jupyter*

#### American Sign Language Vision-to-Text Translator (Full Stack Data Scientist)

CNN Model Trained on a Dataset of 87,000+ Gesture Images – [jamesvdinh.github.io/asl-translator](https://jamesvdinh.github.io/asl-translator)

- ❖ Implemented a feedforward neural network with 128-unit and 256-unit ReLU layers that classified inputs across 29 categories on a live camera feed input; Used binary cross-entropy loss & SGD to achieve an AUC of 0.97
- Python (Pandas, Scikit-learn, Seaborn) / TensorFlow / MediaPipe / OpenCV / JavaScript*

### Skills

- ❖ Python, SQL, TypeScript, JavaScript, React, Vue, NodeJS, PostgreSQL, GraphQL, MongoDB, jQuery, PHP, Azure, C# .NET, Java, Docker, Kubernetes, Flask, HTML, CSS, Git, REST API, Pandas, NumPy, PyTorch, Seaborn, TensorFlow
- ❖ Personal: Lion Dance, LEGOs, Bouldering, Videography, Guitar, Piano, Table Tennis, Chess